Covalent and metallic bonding, and electrolysis

Mark scheme and worked solutions for 20 questions: 7 multiple choice, 13 structured.

1 · Part 3

Paper 1 & 2 | Covalent bonding | [1]

Answer: B

Worked solution

In a covalent bond, each atom contributes one electron to a pair that is shared between the two atoms. No electrons are transferred and no ions are formed.

2 · Part 3

Paper 1 & 2 | Covalent bonding | [1]

Answer: D

Worked solution

Count the shared pairs between the bonded atoms: O–H in water is one pair, N–H in ammonia is one pair, N≡N is three pairs, and O=O is two pairs. Only oxygen has exactly two shared pairs between two atoms.

3 · Part 3

Paper 1 & 2 | Covalent bonding | [1]

Answer: A

Worked solution

Use two clues. A high melting point means a giant structure with strong bonds throughout (A, B or D). Not conducting even when liquid rules out ionic (B) and metallic (D) substances, leaving A.

4 · Part 3

Paper 2 only | Metallic bonding | [1]

Answer: C

Worked solution

In a metal, positive ions are arranged in layers surrounded by delocalised electrons. When a force is applied, the layers slide, but the delocalised electrons still attract all the ions, so the metal changes shape instead of shattering.

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5 · Part 3 Paper 2 only Electrolysis [1]

Answer: A

Worked solution

Opposite charges attract: cations go to the cathode (negative electrode) and anions go to the anode (positive electrode). At the cathode $Pb^{2+} + 2e^- \rightarrow Pb$.

6 · Part 3 Paper 2 only Electrolysis [1]

Answer: C

Worked solution

At the cathode: if the metal is more reactive than hydrogen, hydrogen forms (from H^+ ions in water). At the anode: if the solution is concentrated and halide ions are present, the halogen forms; otherwise oxygen forms. For potassium bromide this gives hydrogen and bromine.

7 · Part 3 Paper 2 only Electrolysis [1]

Answer: B

Worked solution

The cathode is negative and supplies electrons. Positive silver ions gain one electron each: $Ag^+ + e^- \rightarrow Ag$. Remember that oxidation is loss of electrons and reduction is gain of electrons.

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8 · Part 3

Paper 1 & 2 | Covalent bonding | [5]

Mark scheme

(a) **[1]**

- 4 (1)

(b) **[1]**

- 8 (1)

> **ECF** 2 × the number given in (a)

(c) **[2]**

- the nuclei of the atoms are positive and the shared electrons are negative (1)
- the shared pair of electrons is attracted to both nuclei, which holds the two atoms together (1)

> **ALLOW** there is an electrostatic attraction between the shared pair of electrons and the nuclei of both atoms (2)

(d) **[1]**

- it does not conduct electricity (1)

> **IGNORE** reasons

Worked solution

(a) The formula CH_4 shows four hydrogen atoms, each joined to carbon by one single bond, so there are 4 covalent bonds.

(b) Each covalent bond is one shared pair, which is 2 electrons, so 4 × 2 = 8 electrons are shared.

(c) A covalent bond holds atoms together because of electrostatic attraction. The shared pair of electrons is negatively charged and the nucleus of each atom is positively charged. Both nuclei are attracted to the same shared pair, and this pulls the atoms together.

(d) Methane is a covalent compound, and covalent compounds do not usually conduct electricity.

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9 · Part 3

Paper 1 & 2 Covalent bonding [6]

Mark scheme

(a) **[1]**

- the boiling point increases (as the relative molecular mass increases) (1)

(b) **[1]**

- any value in the range 60 °C to 80 °C (1)

(c) **[2]**

- methane, ethane, propane and butane (1)
- their boiling points are below 25 °C (1)

REJECT the first mark if pentane is included or any of the four is missing

(d) **[2]**

- the intermolecular forces (between molecules) are stronger as the relative molecular mass increases (1)
- more (heat) energy is needed to overcome the intermolecular forces (1)

REJECT breaking covalent bonds for the second mark

Worked solution

(a) Reading down the table, the boiling point rises from −162 °C to 36 °C as Mr rises from 16 to 72.

(b) The rise in boiling point gets slightly smaller each time (73, 47, 41, then 37 °C). Continuing the pattern, the next rise is a little over 30 °C, so hexane should boil at about 69 °C.

(c) A substance is a gas at 25 °C if it has already boiled, that is if its boiling point is below 25 °C. Methane, ethane, propane and butane all boil below 25 °C. Pentane boils at 36 °C, so it is still a liquid at 25 °C.

(d) When a simple molecular substance boils, only the weak intermolecular forces between molecules are overcome; the covalent bonds inside the molecules do not break. Larger molecules with a higher Mr have stronger intermolecular forces, so more energy is needed to separate them and the boiling point is higher.

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10 · Part 3 Paper 1 & 2 Covalent bonding [6]

Mark scheme

(a) **[1]**

- it has a high melting point (so it does not melt in the furnace) (1)

ALLOW it is unreactive OR does not react at high temperatures

ALLOW it is a poor conductor of heat

REJECT it is a poor conductor of electricity (not relevant to a furnace lining)

(b) **[5]**

Indicative points, any five for 1 mark each:

- silicon dioxide has a giant covalent structure (1)
- (many) strong covalent bonds must be broken to melt silicon dioxide (1)
- this needs a lot of (heat) energy (1)
- carbon dioxide has a simple molecular structure (1)
- only the weak intermolecular forces between carbon dioxide molecules are overcome (1)
- this needs little energy (1)
- the covalent bonds within carbon dioxide molecules are not broken (1)

REJECT "silicon dioxide has stronger intermolecular forces" for the second mark

REJECT any reference to ions or ionic bonding in either substance (do not award the structure mark for that substance)

Worked solution

(a) The furnace is very hot. Silicon dioxide only melts at 1710 °C, so it stays solid.

(b) Both compounds contain covalent bonds, but their structures are different. Silicon dioxide is a giant covalent structure: every atom is joined to its neighbours by strong covalent bonds throughout the whole lattice. To melt it, many of these strong bonds must be broken, which needs a great deal of energy, so it melts at 1710 °C. Carbon dioxide is made of small separate CO_2 molecules. When it changes state, only the weak intermolecular forces between the molecules are overcome; the strong covalent bonds inside each molecule stay intact. Very little energy is needed, so it changes state at −78 °C.

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11 · Part 3

Paper 1 & 2 | Covalent bonding | [6]

Mark scheme

(a) **[2]**

- (in graphite) each carbon atom is bonded to three other carbon atoms, so one electron per atom is delocalised (1)
- the delocalised electrons can move through the structure (and carry charge) (1)

ALLOW free electrons OR mobile electrons for delocalised electrons

REJECT ions moving (do not award the second mark)

(b) **[4]**

- in diamond, each carbon atom is covalently bonded to four other carbon atoms (1)
- (diamond is a rigid three-dimensional lattice and) many strong covalent bonds would have to be broken to scratch or deform it (1)
- graphite has layers of carbon atoms with weak forces between the layers (1)
- the layers can slide over each other (1)

ALLOW hexagonal layers OR sheets for layers

REJECT "weak covalent bonds between layers"

Worked solution

(a) In graphite, each carbon atom uses only three of its four outer electrons to form covalent bonds. The fourth electron is delocalised: it is not fixed to one atom and can move along the layers. These moving electrons carry the electric current.

(b) In diamond, each carbon atom forms four strong covalent bonds to four other carbon atoms in a rigid three-dimensional network. Nothing can move without breaking many strong bonds, so diamond is very hard and is used on drill tips. In graphite, the atoms are bonded strongly within flat layers, but the forces between the layers are weak. The layers can slide over each other easily, so graphite is soft and slippery.

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12 · Part 3 Paper 1 & 2 Covalent bonding [5]

Mark scheme

(a) **[1]**

- covalent (1)

(b) **[3]**

- C_{60} has (only) weak intermolecular forces between its molecules (1)
- so little force or energy is needed to move the molecules apart or past each other (1)
- diamond is a giant structure in which every atom is held by strong covalent bonds (to four other atoms), which would have to be broken (1)

REJECT "the covalent bonds in C_{60} are weak"

(c) **[1]**

- each carbon atom is (covalently) bonded to three other carbon atoms (1)

ALLOW both contain delocalised electrons

ALLOW both contain rings of six carbon atoms OR hexagons

Worked solution

(a) The carbon atoms inside each C_{60} molecule share pairs of electrons, so they are joined by covalent bonds.

(b) C_{60} is a simple molecular substance. The covalent bonds inside each ball are strong, but the forces between neighbouring balls are weak intermolecular forces. When you press on C_{60}, the molecules are pushed apart without breaking any covalent bonds, so it is soft. Diamond is a giant covalent structure: deforming it would mean breaking many strong covalent bonds, so it is very hard.

(c) In both graphite and C_{60}, each carbon atom is bonded to three other carbon atoms (in diamond it is four).

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13 · Part 3 Paper 1 & 2 Covalent bonding [6]

Mark scheme

(a) **[1]**

- simple molecular (1)

ALLOW molecular OR simple covalent

REJECT giant (covalent)

(b) **[2]**

- (there are only) weak intermolecular forces between I_2 molecules (1)
- little (heat) energy is needed to overcome them (1)

REJECT breaking covalent bonds

(c) **[2]**

- the covalent bonds (within the I_2 molecules) are not broken (when iodine turns into a vapour) (1)
- (so) the covalent bond is stronger than the forces between the molecules (1)

ALLOW the intermolecular forces are weaker than the covalent bond (for the second mark)

(d) **[1]**

- it does not conduct (electricity) (1)

Worked solution

(a) Iodine is made of separate I_2 molecules, so it has a simple molecular structure.

(b) To turn solid iodine into a vapour, the molecules must be separated. The forces between I_2 molecules are weak intermolecular forces, so only a small amount of energy, from gentle warming, is needed.

(c) If the covalent bond inside each I_2 molecule had broken, the vapour would contain single iodine atoms. It contains I_2 molecules, so only the forces between molecules were overcome. This shows that the covalent bond is much stronger than the intermolecular forces.

(d) Iodine is a covalent substance and does not conduct electricity.

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14 · Part 3

Paper 2 only | Metallic bonding | [5]

Mark scheme

(a) **[3]**

- (a giant lattice of) positive (magnesium) ions (1)
- (surrounded by a sea of) delocalised electrons (1)
- (strong) electrostatic attraction between the positive ions and the delocalised electrons (1)

ALLOW Mg^{2+} ions for positive ions

REJECT "atoms share electrons in covalent bonds" (no marks for bonding)

REJECT negative ions

(b) **[2]**

- the (electrostatic) attraction between the positive ions and the delocalised electrons is strong (1)
- a lot of (heat) energy is needed to overcome this attraction (1)

REJECT intermolecular forces OR covalent bonds

Worked solution

(a) Magnesium atoms lose their outer electrons. The atoms become positive ions arranged in a regular lattice, and the lost electrons become delocalised: they are free to move through the whole lattice. Metallic bonding is the strong electrostatic attraction between the positive ions and the delocalised electrons.

(b) To melt magnesium, the ions must be able to move out of their fixed positions. This means overcoming the strong attraction between the ions and the delocalised electrons, which needs a lot of energy, so the melting point is high.

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15 · Part 3

Paper 2 only | Metallic bonding | [6]

Mark scheme

(a) **[2]**

- (copper contains) delocalised electrons (1)
- which move through the metal (when a voltage is applied) (1)

ALLOW free OR mobile electrons

REJECT ions moving

(b) **[2]**

- copper(II) chloride contains ions (1)
- the ions are held in fixed positions in the (solid) lattice and cannot move (1)

REJECT "there are no ions in solid copper(II) chloride"

REJECT "electrons cannot move" alone

(c) **[2]**

- in molten copper(II) chloride the charge is carried by ions, which gain or lose electrons at the electrodes, forming new substances (copper and chlorine) (1)
- in copper wire the charge is carried by delocalised electrons, which flow through the metal without any chemical change (1)

Worked solution

(a) In a metal, the delocalised electrons are not attached to any one ion. When the wire is connected to a power supply, these electrons flow through the metal, which is an electric current.

(b) Copper(II) chloride is made of Cu^{2+} and Cl^{-} ions. In the solid the ions are held in fixed positions in the lattice. They cannot move, and there are no delocalised electrons, so no charge can flow.

(c) In molten copper(II) chloride, the ions can move. The Cu^{2+} ions move to the cathode and gain electrons to form copper; the Cl^{-} ions move to the anode and lose electrons to form chlorine. The compound is broken down: this is electrolysis. In a copper wire, only electrons move. The copper ions stay where they are, so the copper is not changed.

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16 · Part 3 Paper 2 only Electrolysis [6]

Mark scheme

(a) **[1]**

- a negative ion (1)

ALLOW a negatively charged ion

(b) **[2]**

- (sodium chloride solution) contains ions (1)
- the ions are free to move (and carry charge) (1)

REJECT electrons moving (for the second mark)

(c) **[2]**

- glucose is covalent so (the solution) contains no ions (1)
- there are no charged particles that are free to move (1)

ALLOW glucose exists as molecules (for the first mark)

REJECT "glucose molecules are charged"

(d) **[1]**

- only the sodium chloride solution contains ions that move to the electrodes and are discharged, forming gases (1)

ALLOW electrolysis (decomposition) happens only in the sodium chloride solution because only it conducts electricity

Worked solution

(a) An anion is a negative ion (a cation is a positive ion).

(b) Sodium chloride solution contains Na^+ and Cl^- ions. In a solution the ions are free to move, so they carry the charge through the liquid and the lamp lights.

(c) Glucose is a covalent compound: it is made of molecules, not ions. The solution has no charged particles that can move, so no current flows.

(d) Because a current flows through the sodium chloride solution, ions are discharged at the electrodes and gases form. This is electrolysis. No current flows in the glucose solution, so nothing happens at the electrodes.

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17 · Part 3 Paper 2 only Electrolysis [6]

Mark scheme

(a) **[2]**

- cation: Zn^{2+} (1)
- anion: Cl^- (1)

(b) **[2]**

- cathode: $Zn^{2+} + 2e^- \rightarrow Zn$ (1)
- anode: $2Cl^- \rightarrow Cl_2 + 2e^-$ (1)

ALLOW $2Cl^- - 2e^- \rightarrow Cl_2$

ALLOW multiples

IGNORE state symbols

(c) **[1]**

- chloride ions lose electrons (1)

(d) **[1]**

- chlorine is toxic (1)

ALLOW poisonous OR harmful

Worked solution

(a) Zinc forms Zn^{2+} ions and chlorine forms Cl^- ions; the formula $ZnCl_2$ balances one 2+ charge with two 1– charges. The cation (positive ion) is Zn^{2+} and the anion (negative ion) is Cl^-.

(b) At the cathode, Zn^{2+} ions gain two electrons each: $Zn^{2+} + 2e^- \rightarrow Zn$. At the anode, chloride ions each lose one electron and pair up into chlorine molecules: $2Cl^- \rightarrow Cl_2 + 2e^-$. Check that the charges balance on both sides of each equation.

(c) Oxidation is the loss of electrons. At the anode, chloride ions lose electrons, so the reaction is oxidation.

(d) Chlorine gas is toxic, so it must not escape into the room.

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18 · Part 3 Paper 2 only Electrolysis [6]

Mark scheme

(a) **[2]**

- cathode: hydrogen (1)
- anode: chlorine (1)

REJECT formulae of ions

(b) **[2]**

- (hold) damp (blue) litmus paper (in the gas) (1)
- it is bleached OR turns white (1)

ALLOW damp blue litmus turns red then white (2)

ALLOW damp universal indicator paper is bleached (2)

REJECT dry litmus paper (for the first mark)

(c) **[2]**

- the solution contains H^+ ions (from water) as well as Na^+ ions (1)
- sodium is more reactive than hydrogen, so H^+ ions gain electrons (are discharged) instead of Na^+ ions (1)

ALLOW hydrogen is less reactive than sodium

Worked solution

(a) Hydrogen forms at the cathode and chlorine forms at the anode.

(b) Hold a piece of damp blue litmus paper in the gas. Chlorine first turns it red and then bleaches it white.

(c) An aqueous solution contains H^+ and OH^- ions from water as well as the Na^+ and Cl^- ions. Both Na^+ and H^+ are attracted to the cathode. Sodium is more reactive than hydrogen, so its ions hold on to their charge more strongly. The H^+ ions gain electrons instead and hydrogen gas is produced: $2H^+ + 2e^- \rightarrow H_2$.

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19 · Part 3

Paper 2 only | Electrolysis | [7]

Mark scheme

(a) **[2]**

- cathode: hydrogen (1)
- anode: oxygen (1)

(b) **[2]**

- $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$
- correct species: OH^- on the left; O_2, H_2O and electrons on the right (1)
- correctly balanced (1)

ALLOW $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$ (2)

ALLOW multiples

IGNORE state symbols

(c) **[2]**

- the same number of electrons is gained at the cathode as is lost at the anode (1)
- 4 electrons produce 2 molecules OR moles of hydrogen ($2H^+ + 2e^- \rightarrow H_2$, twice) but only 1 molecule OR mole of oxygen, so twice the amount (and volume) of hydrogen forms (1)

ALLOW $2H^+ + 2e^- \rightarrow H_2$ needs 2 electrons per H_2 but 4 electrons are needed per O_2

(d) **[1]**

- some oxygen dissolves in the solution (1)

ALLOW some oxygen reacts with the (graphite) electrode

ALLOW a leak or gas lost while setting up

Worked solution

(a) Sulfuric acid provides H^+ ions, which are discharged at the cathode to form hydrogen. There is no halide ion, so OH^- ions are discharged at the anode to form oxygen.

(b) At the anode: $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$. Check: 4 O and 4 H on each side, and a total charge of 4– on each side.

(c) At the cathode: $2H^+ + 2e^- \rightarrow H_2$. The electrons lost at the anode flow round the circuit and are gained at the cathode, so the same number of electrons is involved at each electrode. For every 4 electrons, one O_2 molecule forms at the anode, but two H_2 molecules form at the cathode ($2 \times 2e^-$). Equal amounts of gases occupy equal volumes, so the volume of hydrogen is twice the volume of oxygen.

(d) The expected volume of oxygen is $40 \div 2 = 20$ cm^3, but only 19 cm^3 was collected. Oxygen is slightly more soluble in water than hydrogen, so a little of it dissolves in the solution.

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20 · Part 3 Paper 2 only Electrolysis [7]

Mark scheme

(a) **[2]**

- independent variable: time (of electrolysis) (1)
- control variable: any one of current, concentration of copper(II) sulfate solution, volume of solution, temperature, size of electrodes, distance between electrodes (1)

REJECT "mass of cathode" for either mark

ALLOW voltage for current

(b) **[1]**

- 0.27 g at 15 minutes (1)

ALLOW the 15-minute result

(c) **[2]**

- mass increases by 0.11 g every 5 minutes OR 0.022 g per minute (1)
- 0.88 g (1)

ALLOW 0.85 g to 0.90 g for the second mark if working is shown

(d) **[2]**

- the blue colour is due to Cu^{2+} ions (1)
- Cu^{2+} ions are removed from the solution as they gain electrons OR are discharged OR form copper at the cathode (1)

REJECT "copper ions are formed" OR "the solution is diluted"

Worked solution

(a) Arjun changes the time for which electrolysis runs, so time is the independent variable. The increase in mass is the dependent variable. To make it a fair test he must keep other factors the same, for example the current.

(b) All the other results go up by 0.11 g every 5 minutes. At 15 minutes the increase should be 0.33 g, but it is 0.27 g, so this result is anomalous.

(c) Ignoring the anomaly, the mass increases by 0.11 g every 5 minutes. 40 minutes is 8 lots of 5 minutes, so the increase is 8 × 0.11 = 0.88 g. Check: 0.022 g per minute × 40 minutes = 0.88 g.

(d) Copper(II) ions, Cu^{2+}, make the solution blue. At the cathode these ions gain electrons and become copper atoms: $Cu^{2+} + 2e^- \rightarrow Cu$. As the experiment goes on, fewer Cu^{2+} ions are left in the solution, so the blue colour fades.

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